EXPERIMENT 2.3

To perform addition and multiplication of two discrete-time sequences and analyze the resulting signal characteristics

Expected Outcomes

- Obtain correct sum and product sequences.
- Preserve the common sample index for aligned sequences.
- Verify representative output samples numerically.
- Interpret how addition and multiplication change sequence values.

Software / Laboratory Environment

MATLAB

Formulae / Analysis

$z[n] = x[n] + y[n]$.
$p[n] = x[n]y[n]$.
Operations are performed for corresponding sample indices.

Percentage error, where applicable = $|\text{Measured} - \text{Theoretical}| / |\text{Theoretical}| \times 100$.

Experimental Procedure

1. Define the sample index n.
1. Enter x[n] and y[n] with aligned indices.
2. Compute the sum sequence.
3. Compute the product sequence using pointwise multiplication.
4. Plot the input and resulting sequences using stem plots.
5. Compare the calculated samples with manual/theoretical results.

Key Conditions & Threshold Values for Analysis

- Both sequences must use a clearly defined and aligned index set.
- Do not confuse pointwise multiplication with convolution.
- Use sufficient plotting range to show all samples.
- Numerical results should match the theoretical calculations.

MATLAB Coding

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% Experiment 2(iii): Addition and Multiplication of Two Sequences

clc; clear; close all;

n = 0:2;

x = [1 2 1];

y = [2 2 1];

zSum = x + y;

zProd = x .* y;

figure;

subplot(2,2,1); stem(n,x,'filled'); grid on;

title('x[n]'); xlabel('n'); ylabel('Amplitude');

subplot(2,2,2); stem(n,y,'filled'); grid on;

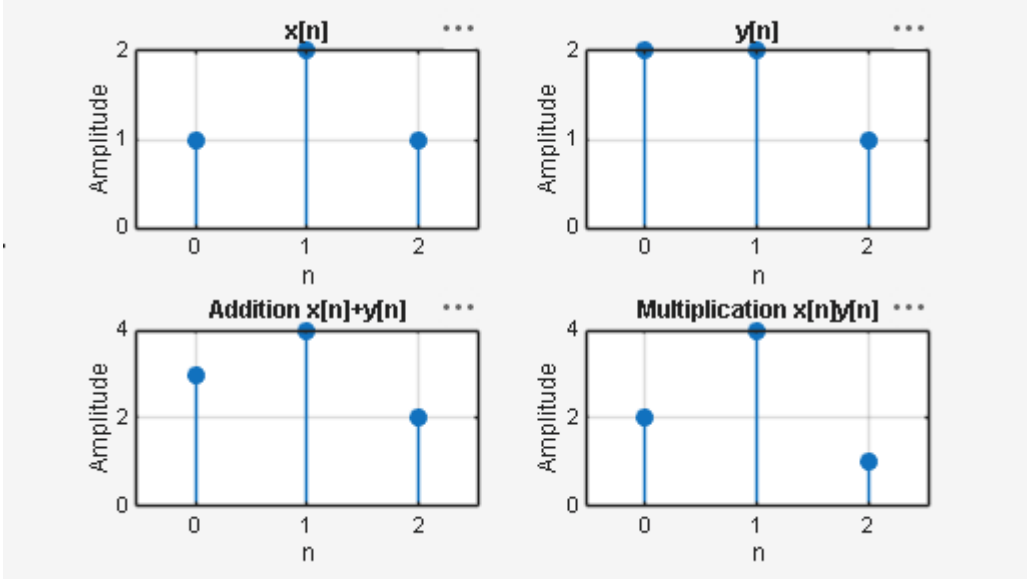
title('y[n]'); xlabel('n'); ylabel('Amplitude');

subplot(2,2,3); stem(n,zSum,'filled'); grid on

title('Addition x[n]+y[n]'); xlabel('n'); ylabel('Amplitude');

subplot(2,2,4); stem(n,zProd,'filled'); grid on;

title('Multiplication x[n]y[n]'); xlabel('n'); ylabel('Amplitude');
```



RESULT:

Amplitude and time scaling, time shifting, time reversal, sequence addition and sequence multiplication were successfully implemented. The original and transformed signals were compared, and the observed waveform changes agreed with the corresponding theoretical signal-operation relationships.

2.

Objective	Assessment Criterion	Excellent – 4	Good – 3	Satisfactory – 2	Needs Improvement – 1	Weightage	Marks(10)	Marks obtained
(i) Perform amplitude scaling and time scaling of continuous-time and discrete-time signals and analyze their effects on signal characteristics.	**Amplitude & Time Scaling Implementation (MATLAB)**	Generates the original and scaled versions correctly for CT and DT with correct index handling. Uses plot for CT and stem for DT, with proper time and index vectors. Original and scaled plots are placed in subplots with matching axes, titles, axis labels and grid.	Generates most scaled signals correctly, with minor errors in scaling factors, index handling, axis limits or labelling.	Generates only basic scaled signals, with noticeable errors in code, time or index vectors, or plots, and needs assistance.	Unable to generate the scaled signals or produce meaningful plots.	**25**	2.5	

(i) Perform amplitude scaling and time scaling of continuous-time and discrete-time signals and analyze their effects on signal characteristics.	**Analysis of Scaling Effects**	Correctly explains that amplitude scaling changes amplitude and energy but not duration or period. Explains that time compression shortens duration and period (T/a) and raises frequency, that expansion does the opposite, and that a < 0 reverses time. Explains that DT scaling discards samples and needs interpolation. Supports every point with waveform evidence.	Explains most effects correctly, with minor errors or limited justification.	Gives basic observations on the effects. The explanation is incomplete and needs guidance.	Unable to analyze or explain the effects of scaling.	**25**	2.5	

(iii) Perform addition and multiplication of two discrete-time sequences and analyzeth e resulting signal characteristics.	**Addition & Multiplication Implementation (MATLAB)**	Adds and multiplies two sequences correctly even when their index ranges differ. Aligns the sequences on a common index vector with zero padding, and uses element-wise operator. Plots both inputs and the results with stem, correct n-axis, titles, axis labels and grid.	Performs both operations correctly, with minor errors in index alignment, padding, operators or labelling.	Performs the operations on simple equal-length sequences only, with noticeable errors in code or plots, and needs assistance.	Unable to perform the operations or produce meaningful stem plots.	**25**	2.5	
(iii) Perform addition and multiplication of two discrete-time sequences and analyze the	**Analysis of Resulting Signal Characteristics**	Correctly determines the amplitude, the length or range of n, and the energy of the sum and	Analyzes most characteristics correctly, with minor errors or limited justification.	Gives basic observations on the resulting signals. The analysis is incomplete and	Unable to analyze or interpret the resulting signals.	**25**	2.5	

resulting signal characteristics.		product. Explains the periodicity of the result (for example, the sum is periodic with period equal to the LCM of the two periods). Explains how multiplication acts as gating or modulation. Interprets the results clearly with waveform evidence.		needs guidance.				
					Total	**100**	**10**	